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Protocol characteristics associated with clinically meaningful 24-hour opioid sparing from perioperative transcutaneous electrical acupoint stimulation and electroacupuncture: a systematic review and meta-regression of randomized controlled trials

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REVIEW TITLE AND BASIC DETAILS

Review title

Protocol characteristics associated with clinically meaningful 24-hour opioid sparing from perioperative transcutaneous electrical acupoint stimulation and electroacupuncture: a systematic review and meta-regression of randomized controlled trials

Review type

None

Condition or domain being studied

Postoperative Pain; Acute Pain; Postoperative Analgesia; Opioids; General Anesthesia; Acupuncture; Electroacupuncture; Acupuncture-like Transcutaneous Electrical Nerve Stimulation; Enhanced Recovery After Surgery

Adults undergoing operative surgery under general anaesthesia. The review evaluates two electrically stimulated acupuncture modalities: (1) transcutaneous electrical acupoint stimulation (TEAS), delivered non-invasively through surface electrodes over recognised body acupuncture points; and (2) needle-based electroacupuncture (EA), in which electrical current is delivered between acupuncture needles inserted at recognised body acupuncture points.

The primary clinical outcome is cumulative systemic opioid consumption during the first 24 postoperative hours, converted to intravenous morphine milligram equivalents (IV MME). Pain at 24 hours is a critical co-outcome used to determine whether opioid reduction occurs without clinically important worsening of analgesia.

Rationale for the review

Postoperative pain is commonly treated with systemic opioids, which may cause nausea and vomiting, sedation, respiratory depression, ileus, delayed mobilisation, and persistent use. Electrical acupuncture techniques are attractive adjuncts because they can be incorporated into multimodal perioperative care and may reduce opioid exposure while maintaining analgesia.

Recent systematic reviews have already addressed whether TEAS, electroacupuncture, or broader acupuncture-related interventions improve postoperative pain and opioid consumption. However, the literature remains highly heterogeneous in treatment timing, number of sessions, cumulative stimulation duration, frequency, intensity, comparator design, surgery type, and background analgesia. Conventional pooled efficacy estimates do not tell clinicians which reproducible protocol characteristics are associated with a clinically meaningful opioid-sparing effect.

TEAS and EA share electrical stimulation but differ in invasiveness, delivery, sensory experience, blinding, and implementation. Pooling them into one overall estimate may obscure clinically important differences. This review will therefore estimate effects separately for TEAS and EA and will use sham-controlled comparisons as the principal evidence base. Usual-care or no-stimulation comparisons will be synthesised separately as supportive evidence.

The review is designed primarily to investigate protocol-level effect modification. Prespecified moderators are intervention timing, single versus repeated treatment, cumulative stimulation duration before the 24-hour endpoint, electrical frequency, and stimulation intensity. Surgical category, regional or neuraxial analgesia, comparator design, and risk of bias will be treated mainly as adjustment or contextual variables rather than as the principal research focus.

Review objectives

The primary objective is to determine which protocol characteristics of perioperative TEAS or EA are associated with greater reduction in cumulative systemic opioid consumption during the first 24 hours after surgery, without clinically important worsening of postoperative pain, in adults undergoing surgery under general anaesthesia.

The review will:

1. Estimate, separately, the sham-controlled effect of TEAS and the sham-controlled effect of EA on cumulative 0-24-hour postoperative opioid consumption expressed as IV MME.

2. Estimate the corresponding effects on pain at rest and pain during movement, approximately 24 hours.

3. Examine, separately within TEAS and EA where data permit, associations between treatment effect and intervention timing, number of treatment sessions, cumulative stimulation duration, electrical frequency, and stimulation intensity.

4. Determine whether the magnitude of opioid reduction reaches a prespecified threshold for clinical importance without clinically important worsening of pain.

5. Provide separate supportive syntheses for TEAS and EA compared with usual care, no stimulation, or other non-sham controls.

6. Assess postoperative opioid consumption beyond 24 hours, intraoperative opioid requirements, pain at additional time points, postoperative nausea and vomiting, rescue analgesia, quality of recovery, functional and gastrointestinal recovery, length of stay, and treatment-related adverse events.

7. Assess risk of bias using Cochrane RoB 2 and certainty of evidence using GRADE.

Keywords

Postoperative opioid consumption; Acute postoperative pain; Postoperative analgesia; Electroacupuncture; Transcutaneous electrical acupoint stimulation; Quality of recovery; Multimodal analgesia; Surgical recovery; Randomized controlled trials; Systematic review; Meta-analysis; Acupuncture; Meta-regression

Country

Sweden

ELIGIBILITY CRITERIA

Population

Included

- Adults aged 18 years or older undergoing an operative surgical procedure under general anaesthesia.
- General anaesthesia alone or general anaesthesia combined with regional or neuraxial anaesthesia or analgesia.
- Any surgical specialty and elective or urgent surgery, provided postoperative opioid consumption or another eligible outcome is reported.
- Mixed-age studies when adult data can be extracted separately or when at least 90% of participants are aged 18 years or older.

Excluded

- Studies involving only children or adolescents.
- Procedures conducted exclusively under local, regional, or neuraxial anaesthesia without general anaesthesia.
- Non-surgical procedures, dental procedures without general anaesthesia, chronic pain populations without an acute perioperative context, and interventions delivered only after recovery from the acute postoperative period.

Intervention(s) or exposure(s)

Included

Acupuncture; Electroacupuncture; Acupuncture-like Transcutaneous Electrical Nerve Stimulation; Perioperative Care; Non-Pharmacological Interventions

1. Transcutaneous electrical acupoint stimulation (TEAS): non-invasive electrical stimulation delivered through surface electrodes positioned over recognised or explicitly defined body acupuncture points, without needle penetration.

2. Electroacupuncture (EA): electrical stimulation delivered through one or more pairs of acupuncture needles inserted at recognised or explicitly defined body acupuncture points.

Eligible terminology may include transcutaneous electrical acupoint stimulation, transcutaneous electric acupoint stimulation, transcutaneous acupoint electrical stimulation, electrical acupoint stimulation, electroacupoint stimulation, acupuncture-like TENS delivered at acupuncture points, electroacupuncture, electric acupuncture, or electrically stimulated acupuncture when the intervention clearly meets the definitions above.

At least one eligible stimulation session must occur before the end of the first 24 postoperative hours. Interventions may be preoperative, intraoperative, postoperative, or delivered across more than one perioperative phase. TEAS or EA may be used alone or as an adjunct to balanced usual multimodal perioperative care.

Trials remain eligible when one or more protocol characteristics are incompletely reported; however, they will contribute only to analyses for which the relevant intervention information is sufficiently available. Protocol characteristics will be extracted according to STRICTA-compatible domains, including modality, timing, number of sessions, session duration, cumulative stimulation duration before 24 hours, acupoints, frequency, waveform, pulse width, intensity strategy, current amplitude, laterality, device, practitioner, and treatment fidelity.

Excluded

- Manual acupuncture without electrical stimulation.
- Dry needling, trigger-point needling, acupuncture-point injection, laser acupuncture, moxibustion, cupping, massage, or herbal interventions.
- Ordinary TENS or percutaneous electrical nerve stimulation delivered only at non-acupuncture locations.
- Auricular acupuncture, battlefield acupuncture, auricular acupressure, electroauriculotherapy, press-tack or intradermal needle interventions, wrist-ankle acupuncture, and buccal acupuncture.
- Electrical stimulation used only for neuromuscular monitoring, nerve localisation, or physiological measurement.
- Mixed interventions in which the independent effect of TEAS or EA cannot be separated.
- Studies in which TEAS and EA are combined within the same intervention arm and their individual contribution cannot be separated.

Comparator(s) or control(s)

Included

PICO tags selected: Sham Acupuncture; Placebo; Usual Care

The principal analyses will use sham-controlled comparisons:

1. TEAS versus inactive or credible sham TEAS, including identical electrode placement with no current, sub-sensory or minimally active current, or non-acupoint sham electrical stimulation.

2. EA versus sham EA, including non-penetrating sham needles, superficial or non-acupoint needling, needles connected to an inactive device, or another credible sham procedure designed to preserve participant blinding.

Usual perioperative care, standard multimodal analgesia, no TEAS or EA, waiting-list control, or attention control will be eligible but analysed separately as supportive comparisons. Active electrical comparators, such as non-acupoint TENS, may be included and reported separately when the contrast is clinically interpretable.

Anaesthetic techniques, regional or neuraxial analgesia, non-opioid analgesics, antiemetics, and postoperative care pathways must be balanced between groups or sufficiently described to assess their potential influence.

Excluded

- Comparators containing an active acupuncture-point intervention that prevents estimation of the independent effect of TEAS or EA.
- Groups receiving substantially different perioperative or analgesic co-interventions without adequate adjustment.
- Comparators that are insufficiently described to classify as sham, usual care, no treatment, or active comparator.

Study design

Only randomized study types will be included.

Included

- Individually randomized parallel-group controlled trials.
- Cluster-randomized trials when appropriate design-adjusted results are available or can be calculated.
- Crossover trials only when an eligible perioperative treatment period can be validly separated and carryover is implausible.

Excluded

- Quasi-randomized studies, non-randomized controlled studies, observational studies, case series, case reports, qualitative studies, reviews, and protocols without results.
- Conference abstracts without sufficient information for eligibility, risk-of-bias assessment, and outcome extraction. Such records may be listed as awaiting classification and investigators may be contacted.

Context

Inpatient, day-surgery, ambulatory surgical, and other perioperative hospital settings in any country or healthcare system will be eligible. Surgical specialties may include orthopaedic, abdominal, gynaecological, urological, thoracic, cardiac, neurological, breast, head and neck, and other operative fields. No restrictions will be applied based on sex, ethnicity, socioeconomic status, or geographic region.

Only studies for which the full-text report is published in English will be eligible for inclusion. No language restriction will be applied at the database-search stage; reports published in languages other than English that are identified by the unrestricted searches will be excluded during study selection on the basis of publication language.

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

19 August 2026

Review timeline

Review start date: 14 August 2026. Review end date: 31 October 2026.

Date of registration in PROSPERO

20 August 2026

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has been written but is not available because:

The protocol is a team-review draft and has not yet been frozen for public release.

SEARCHING AND SCREENING

Search for unpublished studies

Only published studies will be sought.

Main sources that will be searched

The main sources to be searched are *CENTRAL - Cochrane Central Register of Controlled Trials*, *CINAHL - Cumulative Index to Nursing and Allied Health Literature*, *Embase* and *PubMed*.

Search language restrictions

There are no language restrictions.

Search date restrictions

There are no search date restrictions.

Other methods of identifying studies

Other studies will be identified by: *contacting authors or experts*, *looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching)*, *reference list checking (backward citation searching)* and *searching trial or study registers*.

Link to search strategy

A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening

Database-specific TEAS search strategies will combine terminology for transcutaneous electrical stimulation at acupuncture points with surgical or peri-operative concepts and validated randomized-trial terms.

General anaesthesia will be verified during study selection rather than required as a mandatory search concept because exploratory testing showed that a general-anaesthesia search block failed to retrieve known eligible TEAS reports whose anaesthetic technique was incompletely represented in titles, abstracts, or indexing.

Search results will be exported in their original formats, retained unchanged, and deduplicated using documented automated and manual procedures.

No publication-date or language restrictions will be applied to the electronic database searches. This is intended to maximise retrieval and avoid language-related loss of potentially relevant records at the search stage. However, eligibility for the systematic review will be restricted to studies with a full-text report published in English. Reports published only in languages other than English will therefore be excluded during study selection and the reason for exclusion will be recorded as publication language.

Full-text exclusion reasons will be recorded using a prespecified hierarchy, including publication language when the full-text report is not published in English. Excluded full-text reports with reasons will be provided as supplementary material. Searches will be updated before manuscript submission when required.

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment

Risk of bias will be assessed using: *Cochrane RoB-2*

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Risk of bias due to missing results will be assessed by comparing trial registrations, protocols, and publications. Funnel plots and small-study-effect tests will be considered when at least 10 sufficiently similar studies are available. Selective non-reporting will inform sensitivity analyses and GRADE assessments.

Certainty assessment

Certainty of evidence will be assessed using GRADE for each critical and important outcome. Randomized evidence will start at high certainty and may be downgraded for risk of bias, inconsistency, indirectness, imprecision, or publication bias. Judgements will be explained in Summary-of-Findings tables.

OUTCOMES TO BE ANALYSED

Main outcomes

1. Cumulative systemic postoperative opioid consumption from the end of surgery to 24 hours, converted to IV MME using prespecified conversion rules. Programmed but undelivered patient-controlled analgesia doses will not be counted as consumed opioid. The principal effect measure will be mean difference in IV MME.

2. Postoperative pain at approximately 24 hours, assessed separately at rest and during movement where reported. Pain scores will be converted to a 0-10 scale when valid; otherwise standardised mean differences will be used in supportive analyses.

Clinical interpretation: a pooled opioid-sparing effect will be considered clinically meaningful when the point estimate indicates a reduction of at least 10 mg IV MME during 0-24 hours and the corresponding pain evidence does not permit clinically important worsening. Clinically important pain worsening is prespecified as an increase of 1.0 point or more on a 0-10 scale; the upper 95% confidence limit will be examined against this margin. Sensitivity interpretations will use an 8 mg IV MME threshold and, when calculable, a 30% relative opioid reduction.

Additional outcomes

- Cumulative postoperative opioid consumption during 0-48 hours, 0-72 hours, and the longest reported acute postoperative interval.
- Intraoperative opioid consumption, converted to an appropriate common opioid-equivalent metric when valid.
- Pain at rest and during movement at approximately 6, 24, 48, and 72 hours and at the longest reported acute time point.
- Postoperative nausea, vomiting, or composite PONV during 0-24 hours, 0-48 hours, and the longest reported interval.
- Need for rescue analgesia and time to first rescue analgesia.
- Quality of recovery measured using validated instruments such as QoR-15, QoR-40, or QoR-9.
- Patient satisfaction, gastrointestinal recovery, functional recovery, time to mobilisation, and length of hospital stay.
- Persistent postoperative opioid use beyond 30 days, when reported.
- Serious and non-serious adverse events plausibly related to TEAS or EA, including skin reactions, bleeding, infection, electrical discomfort, needle-related adverse effects, and device-related events.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

TEAS and electroacupuncture (EA) will not be combined in a pooled estimate. Analyses will be stratified by modality. Primary comparisons will be TEAS versus credible sham TEAS and EA versus sham EA. Supportive comparisons will evaluate TEAS and EA against usual care, no stimulation, or attention controls. Active comparators will be analysed separately.

When at least two comparable studies are available within a modality–comparator stratum, random-effects meta-analysis will be performed. Continuous outcomes will be pooled using inverse-variance methods and expressed as mean differences or standardised mean differences, as appropriate. Dichotomous outcomes will be pooled as risk ratios. Between-study variance will be estimated using restricted maximum likelihood, with Hartung–Knapp confidence intervals where appropriate.

For multi-arm trials, compatible intervention groups may be combined when justified. Otherwise, shared control groups will be divided or correlations addressed using multivariate methods to avoid double counting. Missing standard deviations may be derived from standard errors, confidence intervals, p values, or other statistics. Median and quantile data will be converted only when assumptions are reasonable and examined in sensitivity analyses.

Statistical heterogeneity will be assessed using forest plots, Cochran's Q, tau-squared, I-squared, and prediction intervals. Clinical heterogeneity will be evaluated before pooling.

Prespecified protocol moderators are intervention timing; single versus repeated treatment before the 24-hour outcome; cumulative active stimulation duration before 24 hours, analysed per 30-minute increment; electrical frequency; and stimulation intensity. Timing will be classified as preoperative only, intraoperative-inclusive, postoperative only, or spanning multiple perioperative phases. Frequency will be grouped as low, intermediate, high, or alternating/mixed. Intensity will be classified as sub-sensory or fixed low current, sensory-level, motor-twitch threshold, maximum tolerated, or insufficiently reported. Actual current amplitude will be analysed only when reporting is comparable.

Meta-regression will be conducted separately for TEAS and EA, without causal claims of superiority. Surgery type, regional or neuraxial analgesia, comparator design, background multimodal analgesia, geographic setting, publication year, and risk of bias will inform adjustment, sensitivity analysis, or interpretation.

Meta-regression will require sufficient study numbers and moderator variation. Normally, at least 10 independent estimates per predictor and four studies per categorical level will be required. Univariable models will be prioritised. Multivariable models will be considered only with at least 20 studies and 10 studies per predictor. Collinearity will be assessed. Results will report coefficients, 95% confidence intervals, residual heterogeneity, and variance explained, and will be interpreted as study-level associations vulnerable to confounding and ecological bias.

If meta-regression is unjustified, protocol characteristics will be synthesised descriptively using structured tables and effect-direction plots. Subgroup analyses will examine sham design, surgical category, regional or neuraxial analgesia, background analgesia balance, and risk of bias.

Sensitivity analyses will restrict or exclude studies based on sham credibility, high risk of bias, imputed or converted data, opioid-conversion uncertainty, unbalanced co-interventions, non-peer-reviewed reports, influential studies, and unresolved cohort overlap. Alternative variance estimators and confidence-interval methods will be tested. Clinical thresholds of 8 mg IV MME and 30% relative opioid reduction will be examined. No network meta-analysis is planned because transitivity and head-to-head evidence are expected to be insufficient.

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification	✓	
Screening search results against inclusion criteria		
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Review affiliation

Lund University

Funding source

Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

Peer review

There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Acupuncture Therapy; Acupuncture Points; Electroacupuncture; Acupuncture Analgesia

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 21 July 2026). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Effectiveness of Medical Acupuncture on Postoperative Opioid Consumption and Pain: A Systematic Review and Meta-Analysis of Randomized Controlled Trials [published 4 September 2025] [CRD420251090635]. The review was judged **not to be similar**
- Efficacy of acupuncture-related interventions for postoperative pain after surgery for perianal abscess in adults: a systematic review and meta-analysis of randomized controlled trials [published 25 April 2026] [CRD420261378143]. The review was judged **not to be similar**
- The opioid-sparing effect of acupuncture after abdominal surgery: a systematic review and meta-analysis protocol [published 26 March 2022] [CRD42022311155]. The review was judged **not to be similar**
- Comparative effectiveness of different acupoint stimulation for relieving acute pain after laparoscopic surgery: a network meta-analysis of randomized trials [published 7 June 2026] [CRD420261417428]. The review was judged **not to be similar**
- Effects of Acupuncture-Related Interventions on Postoperative Gastrointestinal Recovery: A Systematic Review, Network Meta-Analysis, and Pairwise Meta-Analysis [published 16 April 2026] [CRD420261370906]. The review was judged **not to be similar**

PROSPERO version history

- Version 1.0, published 20 Aug 2026

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